INCH-POUND

MIL-PRF-26385H
8 June 2020

SUPERSEDING
MIL-PRF-26385G
24 September 2013

PERFORMANCE SPECIFICATION

HOSE, OXYGEN AND PRESSURIZATION, OZONE RESISTANT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.

This specification covers ozone resistant hoses for use with pressurizing and oxygen breathing systems.

1.2 Size.

This specification covers hoses conforming to the internal diameters and lengths identified in Table I and on Figure 1 (see 6.2).

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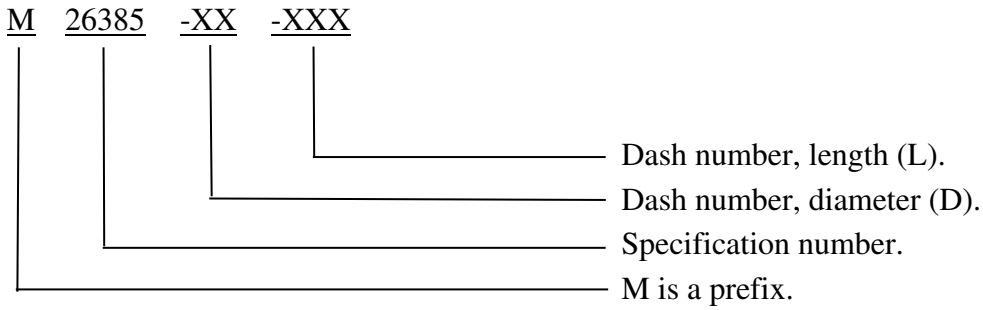
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1.3 Part or identifying number (PIN).

The PINs to be used for the hose acquired to this specification are created as follows (see Figure 1). M26385 dash numbers supersede MS27797 dash numbers.



2. APPLICABLE DOCUMENTS

2.1 General.

The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks..

The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

BB-A-1034	Compressed Air, Breathing

FEDERAL STANDARDS

FED-STD-191	Textile Test Methods

COMMERCIAL ITEM DESCRIPTIONS

A-A-59503	Nitrogen, Technical

DEPARTMENT OF DEFENSE SPECIFICATIONS

AN807	Adapter, Straight, Tube to Hose
MIL-PRF-27210	Oxygen, Aviator's Breathing, Liquid and Gas

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(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.3 Non-Government publications.
The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AEROSPACE INDUSTRIES ASSOCIATION (AIA)

AIA/NAS1922	Clamp, Hose, Band, Tangential Worm, Light Weight

(Copies of this document may be obtained online at http://www.aia-aerospace.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4	Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at http://www.asq.org.)

ASTM INTERNATIONAL

ASTM D1149	Standard Test Methods for Rubber Deterioration – Cracking in an Ozone Controlled Environment

(Copies of this document may be obtained online at http://www.astm.org.)

SAE INTERNATIONAL

SAE AIR1059	Oxygen Cylinder Quality, Serviceability, Maintenance Transfilling, and Marking
SAE AMS-STD-595	Colors Used in Government Procurement
SAE AS1933	Age Controls for Hose Containing Age-Sensitive Elastomeric Material
SAE AS5132	Fitting End – Hose Connection, Design Standard
SAE AS5180	Fitting, Adapter, Bulkhead, Flared to Hose

(Copies of these documents may be obtained online at http://www.sae.org.)

2.4 Order of precedence.
Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article.
When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Recycled, recovered, environmentally preferable, or biobased materials.
Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Materials.
All materials shall be suitably treated to resist corrosion due to electrolytic decomposition and any other atmospheric condition that may be encountered during operational use or storage. The use of toxic chemicals, hazardous substances, or ozone depleting chemicals shall be avoided, whenever feasible.

3.3.1 Ozone resistance.
End cover material, inner liner material, and elastomers shall be ozone resistant.

3.4 Interface.

3.4.1 Color.
The exterior of the hose shall be lusterless green, color 34079 in accordance with SAE AMS-STD-595.

3.4.2 Fittings.
The hose shall be compatible with fittings conforming to SAE AS5180 and SAEAS5132, and with hose clamps conforming to AIA/NAS1922.

3.4.3 Dimensions.
The hose shall conform to the dimensions shown on Figure 1.

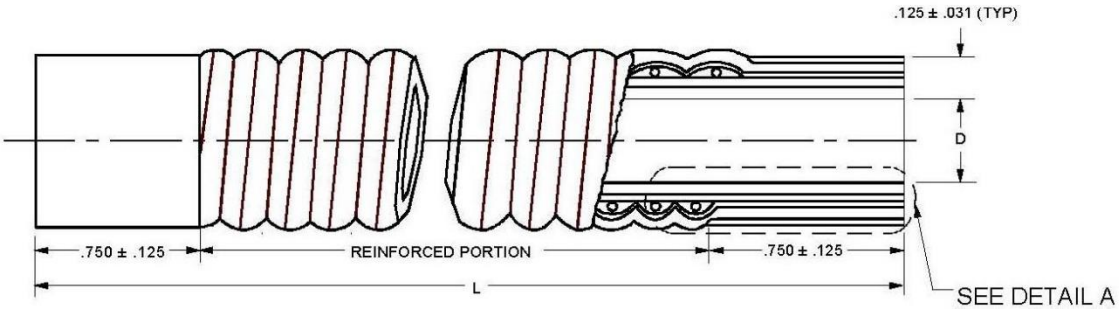


FIGURE 1. Hose.

NOTES:
1. Dimensions are in inches.

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2. Tolerances are as specified.
3. First dash number is inside diameter. The second dash number is length. If second dash number is not listed, length can be specified in other standard lengths.
4. Drawing is not to scale.
5. End notes and dimensions apply to both ends.

3.5 Performance.

3.5.1 Components.

3.5.1.1 Hose reinforcement.
The hose shall be reinforced (see Figure 1).

3.5.1.2 Inner liner.
The hose shall be provided with an inner liner (see Figures 1 and 2).

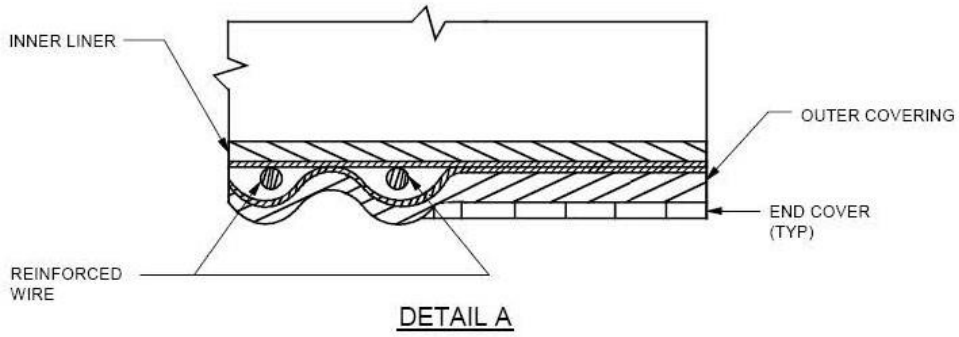


FIGURE 2. Hose end – Detail A.

NOTES:

1. See Table I for hose size determination.
2. Drawing is not to scale.

TABLE I. Hose size.

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M26385 dash number (Note 3)	D □ .031 (Inches)	L + 5%, - 0% (Inches)	Superseding MS27797 dash number
-2	0.125	-	-
-4	0.250	-	-
-6	0.375	-	-
-6-052	0.375	5.2	-6-052
-6-060	0.375	6.0	-6-060
-6-140	0.375	14.0	-6-140
-6-214	0.375	21.4	-6-214
-8	0.500	-	-
-10	0.625	-	-
-10-080	0.625	8.0	-10-080
-10-085	0.625	8.5	-10-085
-10-115	0.625	11.5	-10-115
-10-120	0.625	12.0	-10-120
-10-180	0.625	18.0	-10-180
-10-200	0.625	20.0	-10-200
-10-240	0.625	24.0	-10-240
-12	0.750	-	-

3.5.1.3 End cover.
The hose shall be provided with an end cover 0.750 inches in length on each end (see Figures 1 and 2).

3.5.1.4 Outer covering.
The hose shall be provided with an outer covering (see Figures 1 and 2).

3.5.2 Leakage.
With an internal pressure of 25 psi, the hose shall not leak.

3.5.3 Delamination.
Under a vacuum of 16 inches of mercury, the inner liner shall not delaminate.

3.5.4 Cleanliness.
All surfaces shall be free of visible particulates (50 microns and larger), free of visible fluorinated lubricants, and free of hydrocarbon contamination to a level not greater than 3 milligrams per square foot (mg/ft^2).

3.5.5 Flexibility.
The hose shall be free from any permanent set after being coiled around a 1.5 inch diameter rod and rotated 90 degrees about its longitudinal axis four times.

3.5.6 Elongation.
The hose, when suspended vertically with a 60-pound weight suspended from the lower end, shall

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not elongate more than 2 inches per foot of hose length. Any permanent set shall not exceed 2 percent of the hose length.

3.5.7 Static load.
Under a static load of 225 pounds normal to the diameter of a 4-inch length of the hose, the outside diameter shall not be decreased more than 10 percent of the original value. After removal of the static load, the outside diameter of the hose shall be within 5 percent of the original diameter.

3.5.8 Flexibility endurance.
The hose shall withstand being coiled and uncoiled around a 1.5 inch diameter rod 500 times. The hose shall then withstand being twisted 180 degrees in both directions 500 times.

3.5.9 Abrasion.
The hose shall withstand abrasion occurring during normal usage.

3.5.10 Odor.
The hose shall be odorless.

3.5.11 Low temperature.
The hose shall withstand exposure to -65 ±5 °F for 48 hours.

3.5.12 High temperature.
The hose shall withstand exposure to 160 ±5 °F for 48 hours.

3.5.13 Age.
Hose shall be no more than 12 months old from the cure date to the acceptance date as defined in SAE AS1933.

3.5.14 Weight.
The hose shall not exceed 3.0 ounces per foot of length.

3.5.15 Identification.
The hose shall be permanently and legibly marked with the following data (see 6.2):

a. Nomenclature.
b. Cure date.
c. PIN.
d. National Stock Number.
e. Manufacturer's CAGE code.
f. Manufacturer's part number.

3.6 Interchangeability.
All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable.

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4. VERIFICATION

4.1 Classification of inspections.

The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection.

First article inspection shall be performed on two 12-inch length hoses and shall consist of all tests in section 4.6.

4.3 Conformance inspection.

Conformance inspection shall consist of the individual tests in 4.3.1 and the sampling tests in 4.3.2.

4.3.1 Individual tests.

Each hose shall be subjected to the following tests:

a. Examination (see 4.6.1).
b. Leakage (see 4.6.2).
c. Delamination (see 4.6.3).

4.3.2 Sampling tests.

Sampling tests shall be performed in accordance with ASQ/ANSI Z1.4. Sampling shall begin at the normal inspection level. The sampling tests are as follows:

a. Cleanliness (see 4.6.4).
b. Flexibility (see 4.6.5).
c. Elongation (see 4.6.6).
d. Static load (see 4.6.7).
e. Odor (see 4.6.10).

4.4 Test conditions.

Unless otherwise specified herein, all tests shall be performed under the following test conditions:

a. Ambient atmospheric conditions.
b. Oxygen conforming to Type I of MIL-PRF-27210.

4.5 Requirements cross-reference matrix.

Table II provides a cross-reference matrix of section 3 requirements tested or verified.

TABLE II. Requirements cross-reference matrix.

Requirement	Verification	Requirement	Verification
3.1	4.2	3.5.5	4.6.5, 4.6.8
3.3	4.6.1	3.5.6	4.6.6

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3.3.1	4.6.13	3.5.7	4.6.7
3.4.1	4.6.1	3.5.8	4.6.8
3.4.2	4.6.1, 4.6.2	3.5.9	4.6.9
3.4.3	4.6.1	3.5.10	4.6.10
3.5.1.1	4.6.5, 4.6.6, 4.6.7	3.5.11	4.6.11
3.5.1.2	4.6.1, 4.6.3	3.5.12	4.6.12
3.5.1.3	4.6.1	3.5.13	4.6.14
3.5.1.4	4.6.1	3.5.14	4.6.1
3.5.2	4.6.2	3.5.15	4.6.1
3.5.3	4.6.3	3.6	4.6.1
3.5.4	4.6.4		

4.6 Tests.

4.6.1 Examination.

The hose shall be examined to determine that the materials, interface, inner liner, end cover, outer covering, weight, identification, and interchangeability conform to this specification.

4.6.2 Leakage.

With fittings conforming to SAE AS5180 or SAE AS5132, and secured by a clamp conforming to AIA/NAS1922, the hose shall be subjected to a static pressure of 25 psi for 2 minutes. The hose shall not leak.

4.6.3 Delamination.

With one end of the hose capped, a vacuum equivalent to 16 inches of mercury shall be applied to the hose for 10 minutes. While the vacuum is maintained, the inner liner shall show no evidence of delamination.

4.6.4 Cleanliness.

All surfaces shall be free of visible particulates (50 microns and larger), free of visible fluorinated lubricants, and free of hydrocarbon contamination to a level not greater than 3 mg/ft^2. Cleanliness of the surfaces shall be demonstrated by industrially accepted methods and these cleaning and verification methods shall be identified.

4.6.5 Flexibility.

The hose shall be coiled and uncoiled around a 1.5 inch diameter rod four times; after each coiling and uncoiling the hose shall be turned 90 degrees clockwise about its longitudinal axis. The hose shall have no permanent set.

4.6.6 Elongation.

The hose shall be suspended by one end, vertically, and a 60-pound pull shall be applied for 10 minutes. The length shall increase no more than 16 percent of the hose length. Two minutes after removal of the force, the hose shall have no permanent set in excess of 2 percent of the hose length.

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4.6.7 Static load.
A static load of 225 pounds shall be applied to a 4-inch section of the reinforced portion of the hose, and this force shall be normal to the longitudinal axis. With the load applied, the outside diameter shall not reduce more than 10 percent. Two minutes after the load is removed, the reduction of the outside diameter shall be no more than 5 percent.

4.6.8 Flexibility endurance.
The hose shall be coiled and uncoiled around a 1.5-inch diameter rod 500 times. The hose shall then be twisted 180 degrees in both directions about its longitudinal axis 500 times in each direction. The hose shall then be subjected to the leakage and delamination tests (see 4.6.2 and 4.6.3).

4.6.9 Abrasion.
Two samples of the hose shall be tested in accordance with FED-STD-191, Method 5308, dry conditions. A minimum force of 1 pound shall be maintained between the abradant and hose throughout the test. The reinforced sections of the hose shall withstand 10,000 cycles of abrasion without exposing the reinforced wire.

4.6.10 Odor.
Oxygen conforming to MIL-PRF-27210 shall be flowed through the hose at 10 liters per minute for 2 minutes and then tested for odor in accordance with SAE AIR1059.

4.6.11 Low temperature.
The hose shall be subjected to -65 ±5 °F for 48 hours. After this period and while still at this temperature, the hose shall be subjected to the flexibility test. After returning to room temperature, the hose shall be subjected to the leakage, delamination, and odor tests (see 4.6.2, 4.6.3, and 4.6.10 respectively).

4.6.12 High temperature.
The hose shall be subjected to 160 ±5 °F for 48 hours. While still at 160 °F, the hose shall be subjected to the flexibility test. After returning to room temperature, the hose shall be subjected to the leakage, delamination, and odor tests (see 4.6.2, 4.6.3, and 4.6.10 respectively).

4.6.13 Ozone resistance.
Samples of the end cover, inner liner, and elastomer materials shall be tested in accordance with ASTM D1149. The test samples shall be elongated 20 percent, placed in an ozone free atmosphere for 24 hours, and then subjected to an ozone environment. The temperature shall be 100 ±5 °F, the ozone concentration shall be 120 ±10 parts per million by volume, and the air velocity across the sample shall be at least 2 feet per second. The material shall be exposed to these conditions for 60 minutes. The test slabs shall be examined under 10× magnification for evidence of damage such as blooming, checking, or cracking.

4.6.14 Age.
The age of the hose shall be verified to be no more than 12 months from cure date to the acceptance date in accordance with SAE AS1933.

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5. PACKAGING

5.1 Packaging.

For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use.

The hoses covered by this specification are intended for use with aircraft oxygen and pressurizing systems. The hoses supply air or oxygen for breathing or pressurizing purposes when used with altitude suits, anti-G suits, and ejection seat or escape capsule equipment.

6.2 Acquisition requirements.

Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Size (see 1.2).
c. When first article is required (see 3.1 and 4.2)
d. Item identification (see 3.5.15 items a, c, and d).
e. The requirement for the vendor to identify proposed cleaning and verification methods (see 4.6.4).
f. Data required.
g. Packaging requirements (see 5.1).

6.3 Subject term (key word) listing.

Altitude suit
Anti-G suit
Ejection seat
Escape capsule

6.4 Changes from previous issue.

Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

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Air Force - 71

Review activities:
Air Force - 11
DLA - CC

Preparing activity:
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